

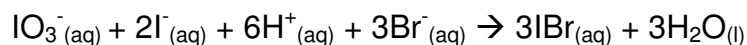
Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1** A sample of 3.42g of aluminium(III) sulphate is dissolved in water to give a 500 cm³ of aqueous solution. What is the concentration of the sulphate ions?

- A** $1 \times 10^{-3} \text{ mol dm}^{-3}$
- B** $2 \times 10^{-2} \text{ mol dm}^{-3}$
- C** $3 \times 10^{-2} \text{ mol dm}^{-3}$
- D** $6 \times 10^{-2} \text{ mol dm}^{-3}$

- 2** Which of the following statements about the reaction below is correct?



- A** The oxidation number of iodine in $\text{IO}_3^- (\text{aq})$ changes from +5 to +1.
- B** The oxidation number of iodine in $\text{IO}_3^- (\text{aq})$ changes from +5 to -3.
- C** The oxidation number of bromine changes from -1 to -2.
- D** The oxidation number of bromine changes from -1 to +1.

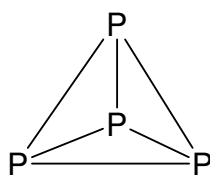
- 3** In which one of the following reactions does hydrogen behave as an oxidising agent?

- A** $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$
- B** $2\text{Na} + \text{H}_2 \rightarrow 2\text{NaH}$
- C** $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
- D** $\text{C}_2\text{H}_5\text{CHO} + \text{H}_2 \rightarrow \text{C}_2\text{H}_5\text{CH}_2\text{OH}$

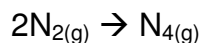
- 4 In a sample of chlorine gas containing ^{35}Cl and ^{37}Cl , masses of particles could be
- A 70, 71, 72
 - B 70, 72, 73
 - C 71, 73, 74
 - D 70, 72, 74
- 5 Which one of the following corresponds to the configuration of the three electrons of highest energy for one of the elements in Group III in the ground state?
- A $1s^2 2s^1$
 - B $1s^1 2s^1 3p^1$
 - C $3p^3$
 - D $4s^2 4p^1$
- 6 Why is the boiling point of methane greater than that of neon?
- A The molecule of methane is polar, but the atom of neon is not.
 - B A molecule of methane has more electrons than an atom of neon.
 - C Molecules of methane have stronger intermolecular forces than those of neon.
 - D Molecules of methane form hydrogen bonds, but atoms of neon do not.

- 7 Trichloromethane, CHCl_3 , dissolves in propanone, CH_3COCH_3 . Which of the following provides the **most** appropriate explanation for the above statement?
- A CHCl_3 and CH_3COCH_3 are polar molecules and they form permanent dipole-permanent dipole attractions with each other.
 - B CHCl_3 and CH_3COCH_3 are non-polar molecules and they form induced dipole-induced dipole attractions with each other.
 - C The enthalpy change of reaction is exothermic.
 - D Both are organic molecules.
- 8 Which one of the following sets of solid substances includes a giant metallic structure, a giant covalent structure and a simple molecular structure?
- A Al, C, ceramic
 - B Pb, Al_2O_3 , I_2
 - C Be, Si, SO_3
 - D Ni, SiO_2 , KI
- 9 Which of the following processes is endothermic?
- A The electrolysis of water.
 - B Neutralisation of sulphuric acid.
 - C The freezing of water.
 - D The condensation of steam.

- 10 Nitrogen exists as the molecule $\text{N}\equiv\text{N}$, while phosphorus as the molecule shown.



Imagine that nitrogen were to form a similar molecule N₄.



By considering the bonds broken and the bonds formed, what would be the value of ΔH for the above reaction?

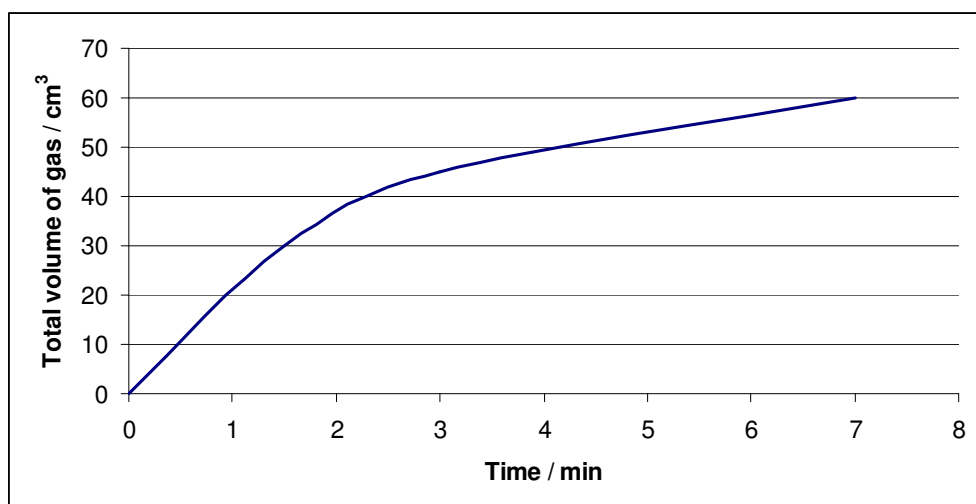
[Bond energies: $\text{N}-\text{N}$, 160 kJ mol^{-1} , $\text{N}\equiv\text{N}$, 994 kJ mol^{-1}]

- A** 1028 kJ mol^{-1}
- B** 1348 kJ mol^{-1}
- C** 1954 kJ mol^{-1}
- D** 2628 kJ mol^{-1}
- 11 Given the following reaction in a vessel:
- $$\text{A}_{(\text{g})} + \text{B}_{(\text{g})} \rightleftharpoons 2\text{C}_{(\text{g})} + \text{D}_{(\text{g})} \quad \Delta H^\theta < 0$$
- Which of the following conditions increase **BOTH** the rate of reaction and yield of the products?
- A** Add a suitable catalyst and add a suitable solvent to dissolve C.
- B** Increase temperature and increase pressure.
- C** Decrease the temperature and increase the pressure.
- D** Decrease the volume of the vessel.

12 A solution of ammonia in water contains

- A** simple molecules only.
- B** simple molecules and hydrogen bonded molecules only.
- C** simple molecules, hydrogen bonded molecules and ions.
- D** ions only.

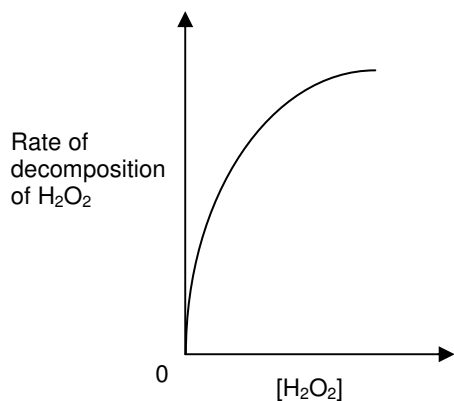
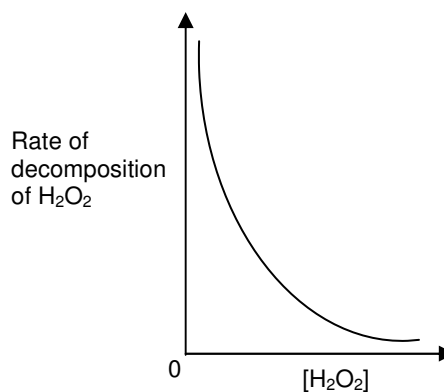
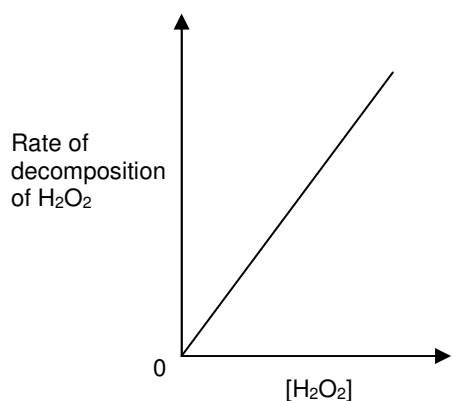
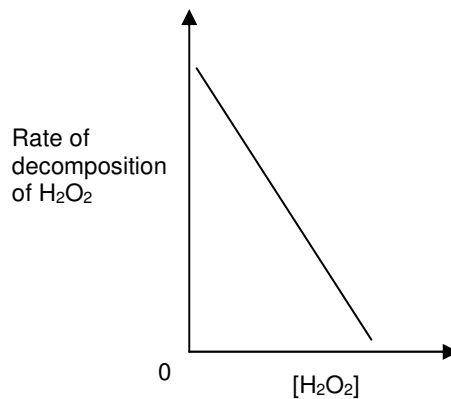
13 A student investigated the reaction rate between sodium and ethanol. A freshly cut piece of sodium was weighed and added to a large excess of ethanol. The total volume of gas liberated was recorded every minute. The results were shown in the graph below.



The half-life(in minutes) of the reaction is

- A** 0.5
- B** 0.75
- C** 1.5
- D** 3.5

- 14 Which one of the following graphs would confirm that the decomposition of hydrogen peroxide was first order with respect to the concentration of hydrogen peroxide?

A**C****B****D**

- 15 Which of the following will completely dissolve in water to form an acidic solution with the evolution of HCl gas?

A Cl_2 **B** AlCl_3 **C** SiCl_4 **D** MgCl_2

16 An element has the following properties:

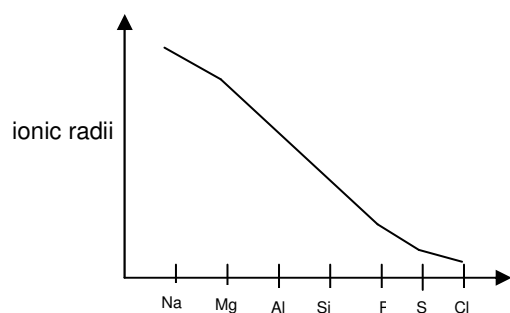
- i) It has a melting point of 1535°C .
- ii) Its density is 7.86 g cm^{-3} .
- iii) It can exist in 2 oxidation states.

What is the element?

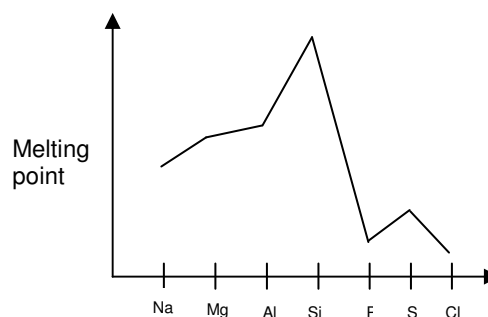
- A** Na
- B** Fe
- C** Si
- D** Hg

17 Which of the following graphs shows the correct trend in physical property of the Period III elements?

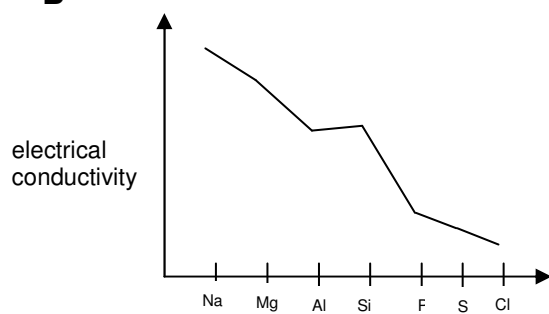
A



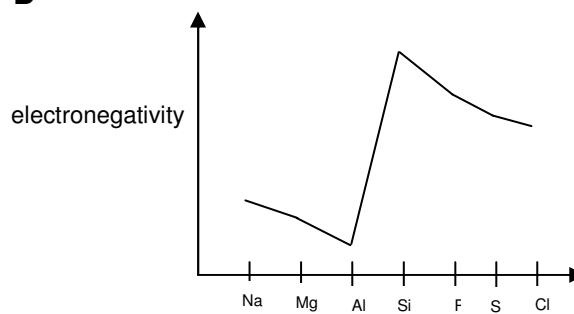
C



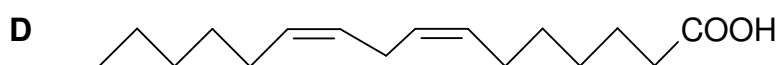
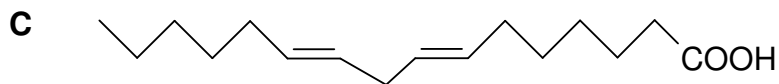
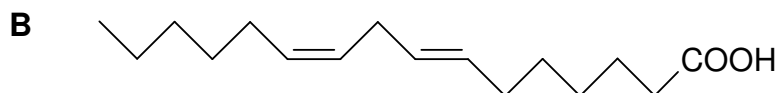
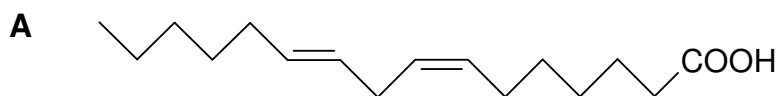
B



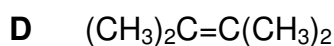
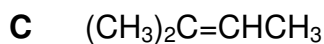
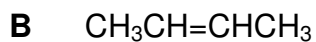
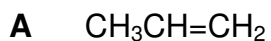
D



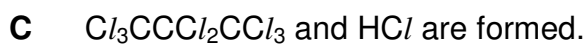
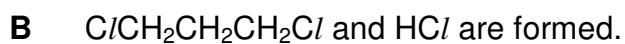
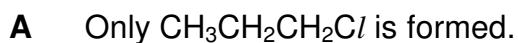
18 Which of the following formulae contains a *trans-trans* arrangement?



19 Which hydrocarbon, on treatment with hot acidified potassium manganate (VII), would give ethanoic acid as the only product?



20 What happens when one mole of propane is mixed in the dark at room temperature with 4 moles of chlorine?



- 21** Dichlorodifluoromethane, CCl_2F_2 , is widely used in aerosol propellants and as a refrigerant.

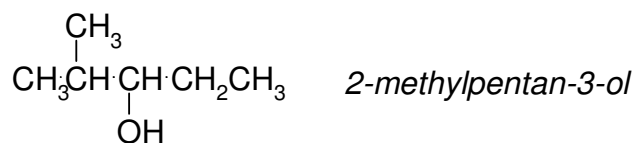
Which statement helps to explain why dichlorodifluoromethane is chemically inert?

- A** The carbon-fluorine bond energy is large.
- B** The carbon-fluorine bond has a low polarity.
- C** Fluorine is highly electronegative.
- D** Fluorine compounds are non-flammable.

- 22** Why does the reaction $\text{CH}_3\text{CH}_2\text{X} + \text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{X}^-$ take place more rapidly in aqueous solution when X is Br than when X is Cl?

- A** The Br^- is a stronger nucleophile than Cl^- .
- B** The Br^- is less hydrated in solution than Cl^- .
- C** The C-Cl bond is more polar than C-Br bond.
- D** The C-Br bond is weaker than the C-Cl bond.

- 23 How many alkenes (including geometric isomers) can be obtained when 2-methylpentan-3-ol undergoes dehydration?



- A 2
B 3
C 4
D 5

- 24 Pentan-2-one is treated with cold hydrogen cyanide in the presence of traces of sodium cyanide. The product **X** of this reaction is then boiled with hydrochloric acid to give **Y**. What are **X** and **Y** likely to be?

- | | X | Y |
|----------|--|--|
| A | $ \begin{array}{c} \text{H} \quad \text{OH} \quad \text{H} \\ \quad \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \quad \quad \\ \text{H} \quad \text{CN} \quad \text{H} \end{array} $ | $ \begin{array}{c} \text{H} \quad \text{OH} \quad \text{H} \\ \quad \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}-\text{CH}_3 \\ \quad \\ \text{H} \quad \text{COOH} \end{array} $ |
| B | $ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{OH} \\ \quad \quad \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{CN} \end{array} $ | $ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{OH} \\ \quad \quad \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{COOH} \end{array} $ |
| C | $ \begin{array}{c} \text{H} \quad \text{OH} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \quad \\ \text{H} \quad \text{CN} \end{array} $ | $ \begin{array}{c} \text{H} \quad \text{OH} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \quad \\ \text{H} \quad \text{COOH} \end{array} $ |
| D | $ \begin{array}{c} \text{H} \quad \text{H} \quad \text{OH} \\ \quad \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{CN} \end{array} $ | $ \begin{array}{c} \text{H} \quad \text{H} \quad \text{OH} \\ \quad \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{COOH} \end{array} $ |

25 Which one of the following is true of the product formed when ethanol is distilled with acidified aqueous potassium dichromate (VI)?

- A** It can be reduced to ethanoic acid.
- B** It undergoes reaction with PCl_5 .
- C** It undergoes nucleophilic addition reactions.
- D** It would **not** give a precipitate with Fehling's solution.

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to place a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Which of the following compounds contain(s) delocalised electrons?

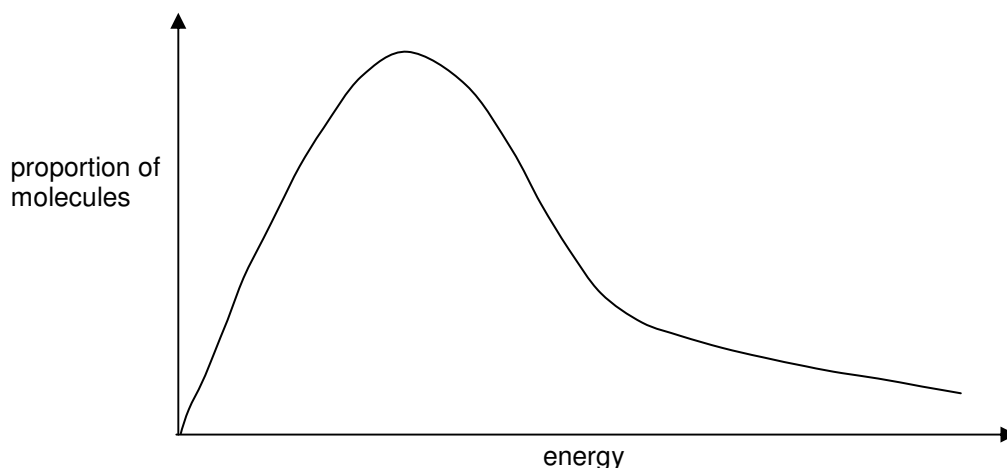
1 cyclohexene

2 graphite

3 sodium

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 27** The diagram represents the Boltzmann distribution of molecular energies at a given temperature.



As temperature increases, which of the following statements is correct?

- 1 The proportion of molecules with energies above a given value increases.
- 2 At all energies, the proportion of molecules of a particular energy increases.
- 3 The activation energy is lowered.

- 28** In which sequence(s) are the molecules quoted in order of increasing bond angle within the molecule?

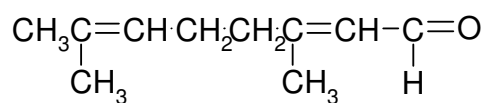
- | | | | |
|---|------------------|-----------------|-----------------|
| 1 | H ₂ O | NH ₃ | CH ₄ |
| 2 | H ₂ O | SF ₆ | BF ₃ |
| 3 | CH ₄ | CO ₂ | SF ₆ |

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

29 Which of the following compounds does **NOT** have the same empirical formula as the rest?

- 1 propanone
- 2 ethanal
- 3 butanoic acid

30 Citral, which occurs in lemongrass oil, has the following structural formula.



From this it can be deduced that citral

- 1 decolorises dilute aqueous potassium manganate(VII).
- 2 gives a silver mirror with aqueous diammine silver(I) ions (Tollen's reagent).
- 3 gives a sweet smelling oil with ethanoic acid, when heated under reflux with a drop of concentrated sulphuric acid.

